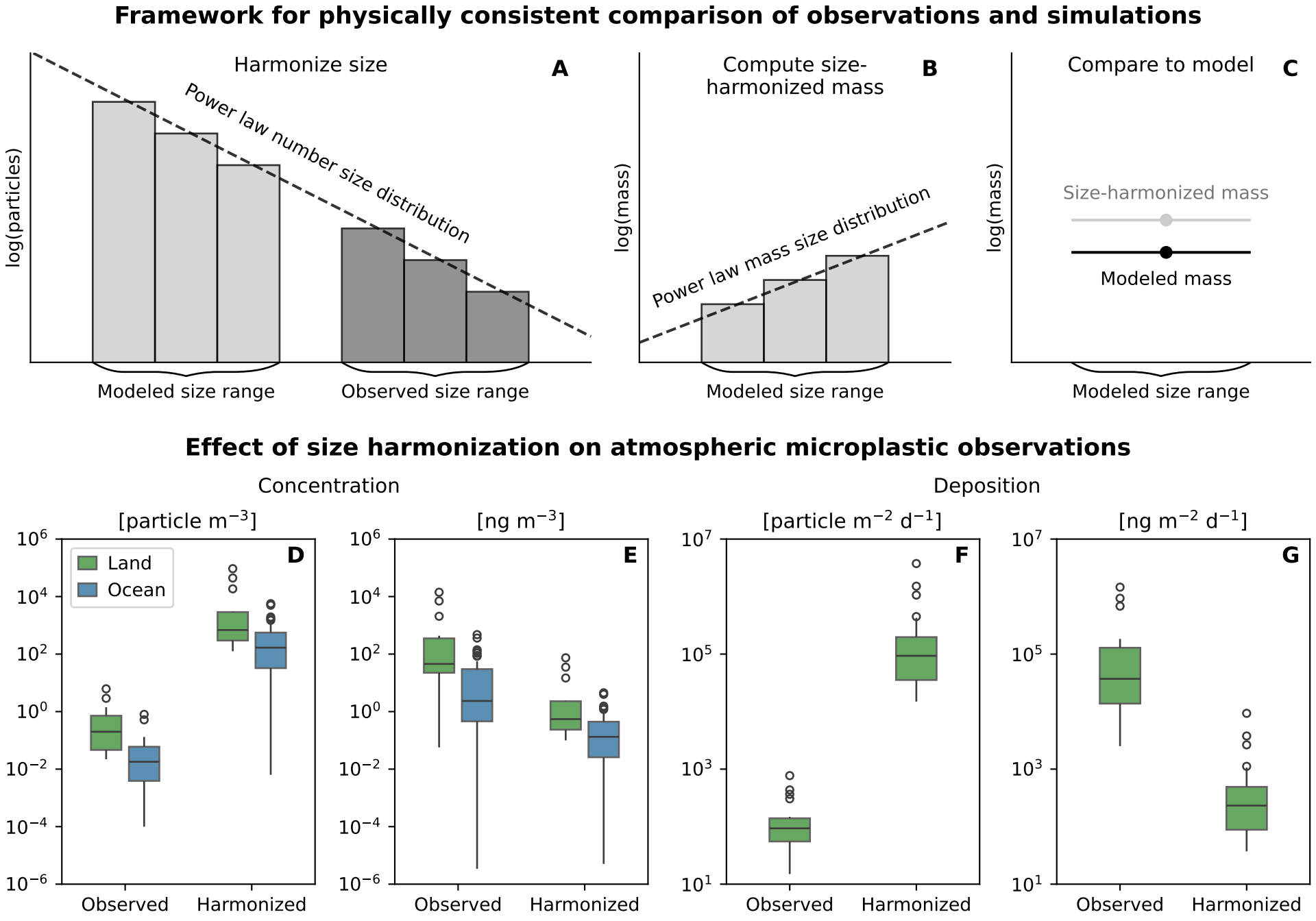
Framework for physically consistent comparison of observations and simulations
Harmonize size
A
Power law number size distribution
log(particles)
Modeled size range
Observed size range
Compute size-harmonized mass
B
log(mass)
Power law mass size distribution
Modeled size range
Compare to model
C
log(mass)
Size-harmonized mass
Modeled mass
Modeled size range
Effect of size harmonization on atmospheric microplastic observations
Concentration
Deposition
[particle m^{-3}]
[ng m^{-3}]
[particle m^{-2} d^{-1}]
[ng m^{-2} d^{-1}]
Land
Ocean
D
E
F
G
Observed
Harmonized